

# Computational Financial Modelling

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## Assignment-2

### IBEX-35 Quantitative Analysis

This project performs a **quantitative analysis of IBEX-35 stocks** using historical market data from **June 2010 to December 2025**.

The workflow includes **data cleaning, statistical analysis, correlation analysis, portfolio simulation, and Markowitz portfolio optimization**.

The analysis is implemented using **Python (pandas, NumPy, SciPy, seaborn, matplotlib)**.

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## Project Structure

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- |— Assignment-2-CFM.pdf
- |— IBEX35/
- |— IBEX35.csv
- |— Q1.py
- |— Q2.py
- |— Q2-Results/
- |— Q3.py
- |— Q3-Results/
- |— Q4.py
- |— Q4-Results/
- |— Q5.py
- |— Q5-Results/
- |— Q6.py
- |— Q6-Results/
- |— Q7.py
- |— Report.pdf
- |— SplitsDivCombo.csv
- |— structure.txt

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# Main Files

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## Assignment-2-CFM.pdf

The original assignment document describing the tasks required for the project.

## Report.pdf

Final compiled report containing methodology, plots, and analysis.

## structure.txt

Text file showing the project directory structure.

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# Data Sources

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## IBEX35.csv

Contains historical data for the IBEX-35 index.

## SplitsDivCombo.csv

Dataset containing stock split and dividend adjustment information.

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# IBEX35 Data Directory

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IBEX35/

├── Index.csv

├── MasterPriceDB.csv

└── results/2010\_6\_to\_2025\_12/

## Index.csv

Historical index values for the IBEX-35 benchmark.

## MasterPriceDB.csv

Combined master dataset containing historical prices for all stocks.

## results/2010\_6\_to\_2025\_12/

Folder having the stock price data for the stocks to be analyzed (individual stocks).

Each CSV file contains:

- Date
- Open
- High
- Low
- Close
- Volume
- Ticker
- Company

Example:

- Acciona.csv
- ACS.csv
- BBVA.csv
- Iberdrola.csv
- Inditex.csv
- ...

Total stocks analysed: **19 IBEX-35 companies**

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# Python Scripts

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Each script corresponds to a specific assignment task (question wise).

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## Q1 – Data Verification and Cleaning

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## Q1.py

Performs:

- Missing value detection
- Duplicate record checks
- Date ordering verification
- Trading day continuity analysis
- Preliminary validation of dataset integrity

Purpose:

Ensure that the dataset is **clean and consistent before performing analysis**.

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## Q2 – Exploratory Data Analysis

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### Q2.py

Tasks:

- Load and combine all stock data
- Compute **daily returns**
- Plot **stock price time series**
- Plot **daily return series**
- Compute descriptive statistics

### Output Directory

Q2-Results/

Contains:

Stocks/

Stock price time series plots for each company.

Example:

- AccionaStock.png
- InditexStock.png
- RepsolStock.png

## Returns/

Daily return time series plots.

Example:

- ACSDailyReturns.png
- BBVADailyReturns.png

## stats.txt

Descriptive statistics for each stock:

- Mean return
- Standard deviation
- Skewness
- Kurtosis
- Minimum return
- Maximum return

## tailEvents.csv

Extreme return observations detected in the dataset.

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# Q3 – Distribution and Normality Analysis

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## Q3.py

Performs statistical analysis of return distributions.

Tasks:

- Generate **histograms of returns**
- Generate **Q-Q plots**
- Perform **normality tests**

Statistical tests used:

- Jarque-Bera test
- Shapiro-Wilk test

## Output Directory

## Q3-Results/

Contains:

### Histograms/

Histogram plots of return distributions.

Example:

- GrifolsHist.png
- BBVAHist.png

### QQPlots/

Quantile-Quantile plots comparing empirical returns to a normal distribution.

Example:

- InditexQQ.png
- IberdrolaQQ.png

### All/

Combined distribution analysis for all stocks:

- allHist.png
- allQQ.png
- allNorms.txt

### Individual.csv

Table containing normality test results for each stock.

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## Q4 – Correlation Analysis

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### Q4.py

Computes relationships between stock returns.

Tasks:

- Construct correlation matrix

- Identify highly correlated stock pairs
- Generate correlation heatmap

## Output Directory

Q4-Results/

Files:

- `CorrHeatmap.png` – visual heatmap of stock correlations
  - `CorrMatrix.csv` – full correlation matrix
  - `CorrPairs.txt` – list of most and least correlated pairs
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# Q5 – Random Portfolio Simulation

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Q5.py

Simulates random portfolios using two selected stocks.

Steps:

1. Generate **5000 random portfolios**
2. Assign random portfolio weights
3. Compute portfolio return
4. Compute portfolio risk (standard deviation)

## Output Directory

Q5-Results/

Files:

- `randPortfolios.csv` – portfolio return and risk values
- `RiskvsReturn.png` – risk-return scatter plot

This plot visualizes the **feasible portfolio region**.

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# Q6 – Markowitz Portfolio Optimization

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## Q6.py

Implements **Modern Portfolio Theory**.

Tasks:

- Compute expected returns
- Compute covariance matrix
- Solve **minimum variance portfolio**
- Generate **efficient frontier**

## Output Directory

Q6-Results/

Files:

- `effFront.png` – efficient frontier plot
  - `minVarWgt.csv` – optimal portfolio weights
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# Q7 – Observations

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## Q7.py

Generates summary observations regarding:

- return distributions
- portfolio feasibility region
- efficient frontier interpretation
- minimum variance portfolio insights

Full explanation is available in the main report ( `Report.pdf` ).

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# Key Concepts Covered

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The project demonstrates the following quantitative finance concepts:

- Financial return distributions
- Heavy-tailed behaviour of markets

- Correlation and diversification
  - Portfolio risk measurement
  - Random portfolio generation
  - Efficient frontier
  - Minimum variance portfolio
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## Final Outcome

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This project shows the basic python implementation of the "Markowitz Portfolio Theory" along with basic data tuning and cleaning for the further analysis.